

Adder, RAM, FIFO 검증 보고서

기본 RTL 블록을 대상으로 UVM testbench 구조, factory 기반 class 생성 흐름, sequence/scoreboard/coverage 구성, Verdi 결과 화면을 정리한다.

3 DUT

Adder / RAM / FIFO

UVM

test, env, agent, sequence

Checker

scoreboard + coverage + SVA

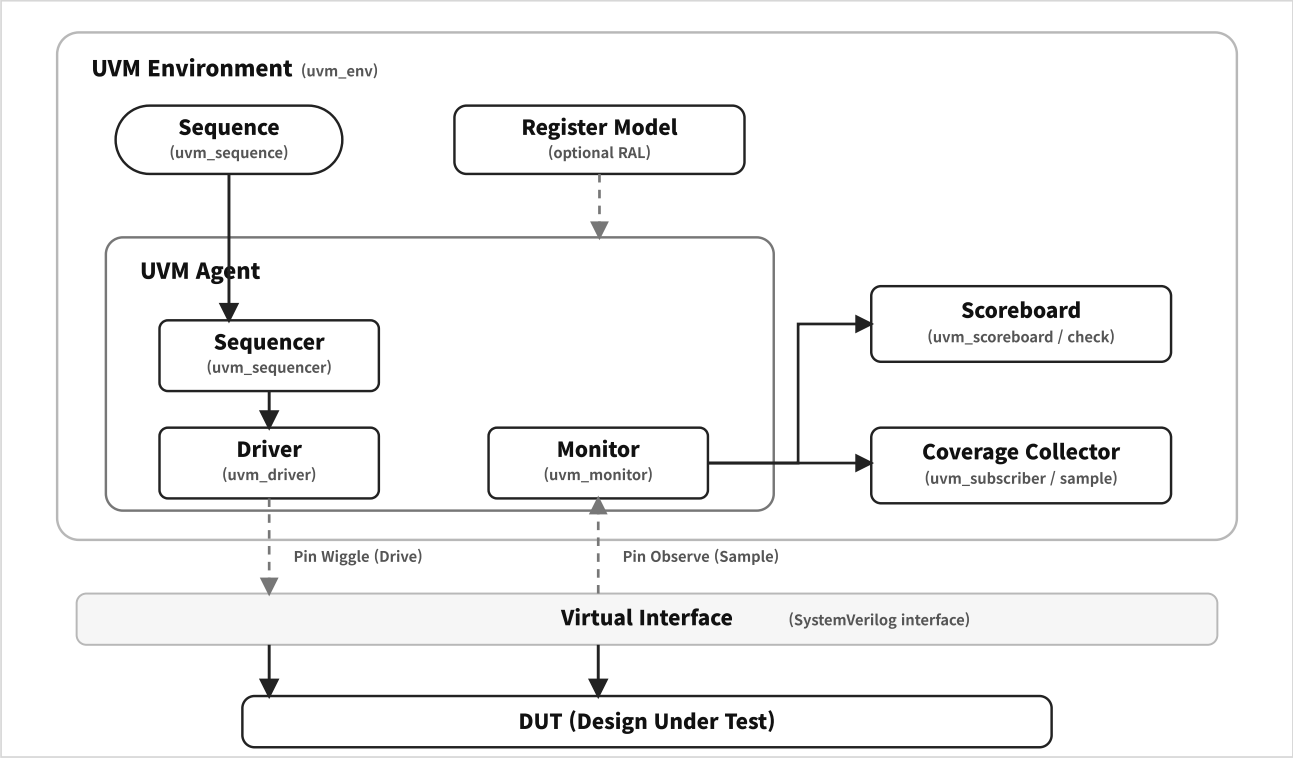
Evidence

Verdi waveform / result / coverage

목차

01	프로젝트 개요 실습 목적, 검증 대상, 공통 UVM 검증 방식	03
02	공통 UVM 구조 top class, factory 생성, build/connect/run 흐름	04-05
03	Adder 검증 RTL 설명, class 구성, sequence/coverage, 결과 화면	06-09
04	RAM 검증 RTL 설명, reference memory, coverage/SVA, 결과 화면	10-13
05	FIFO 검증 RTL 설명, queue model, full/empty coverage, 결과 화면	14-17
06	비교 및 결론 블록별 reference model, timing, coverage 관점 비교	18-19

프로젝트 개요



UVM environment / agent / analysis / virtual interface 구조

검증 대상

DUT	검증 중심
Adder	조합 덧셈, carry bit
RAM	synchronous read/write
FIFO	ordering, full/empty/count

검증 방식

- sequence가 directed/random transaction 생성
- driver가 virtual interface로 DUT 구동
- monitor transaction을 scoreboard와 coverage로 전달
- RAM/FIFO는 protocol assertion 추가

공통 UVM Factory / Top Class 구조

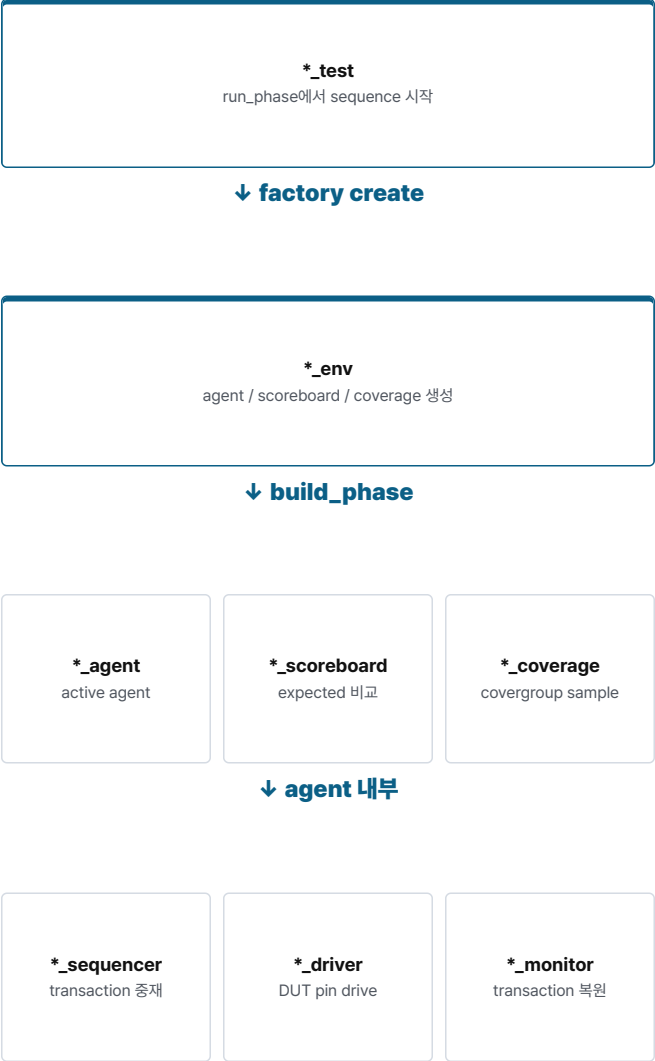
factory 생성 흐름

```
env = *_env::type_id::create("env", this);
agt = *_agent::type_id::create("agt", this);
scb = *_scoreboard::type_id::create("scb", this);
cov = *_coverage::type_id::create("cov", this);

seq = *_sequence::type_id::create("seq");
seq.start(env.agt.sequencer);
```

top class 역할

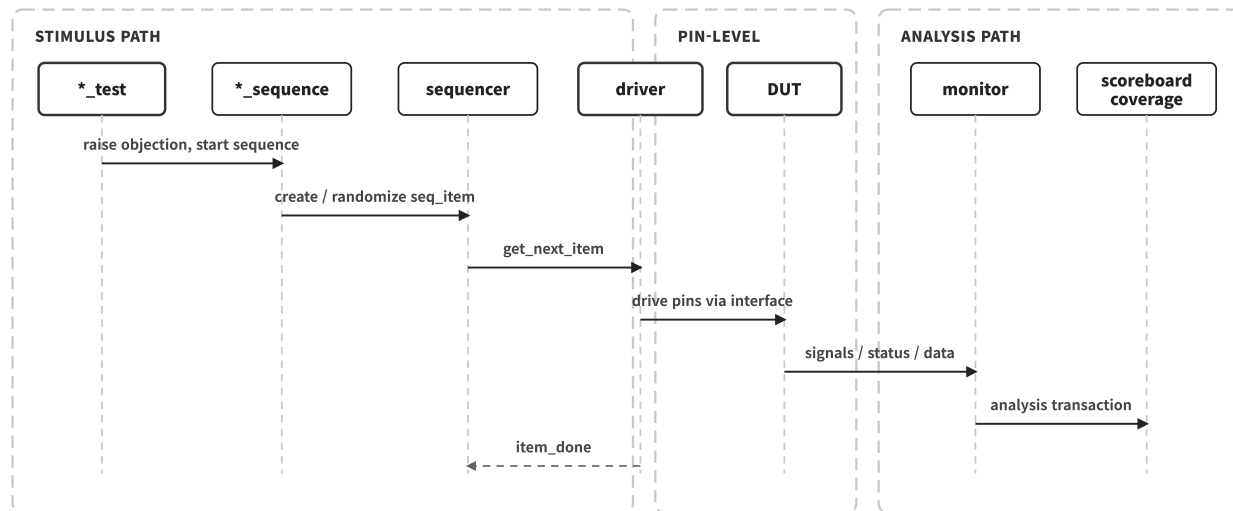
Class	역할
test	env 생성, sequence 실행, objection 제어
env	agent, scoreboard, coverage 연결
agent	sequencer, driver, monitor 구성



공통 Transaction 흐름

Figure 2. Common UVM Transaction Sequence

Stimulus path, DUT boundary, and analysis path shown as lifelines instead of process boxes.



Adder, RAM, and FIFO keep this topology; only timing, fields, reference model, and coverage bins differ.

sequence → driver → DUT → monitor → scoreboard/coverage

Stimulus path

- test가 sequence 시작
- sequence가 seq_item 생성 및 randomize
- sequencer가 driver에 item 전달

Analysis path

- monitor가 DUT signal을 transaction으로 복원
- scoreboard가 reference model과 비교
- coverage가 command/data/status bin sample

Adder RTL

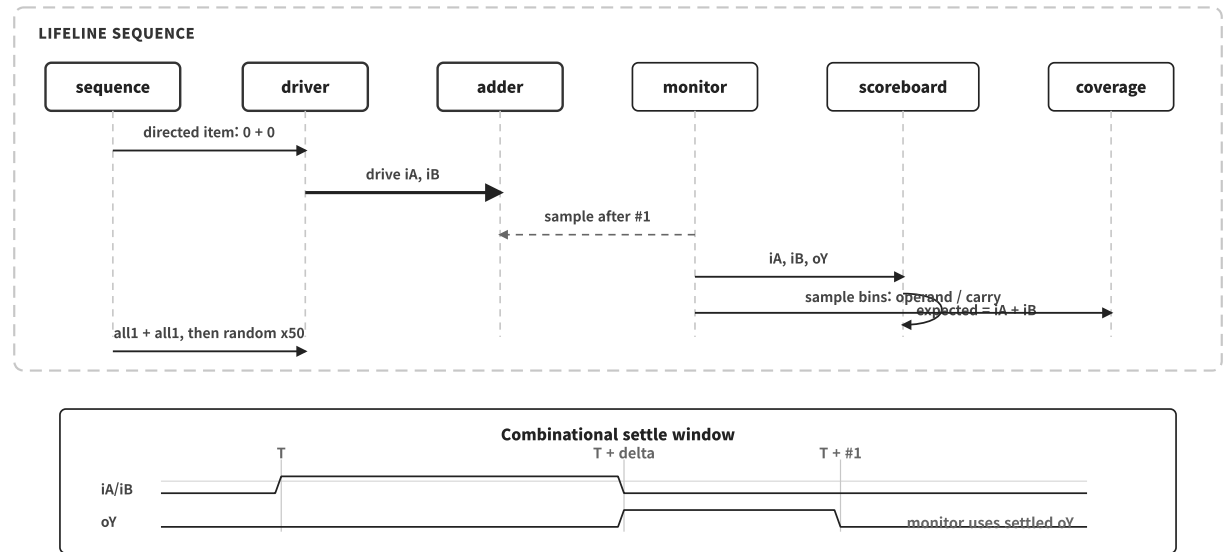
모듈 동작

Signal	설명
iA, iB	DATA_WIDTH bit 입력 operand
oY	DATA_WIDTH+1 bit 출력, MSB는 carry

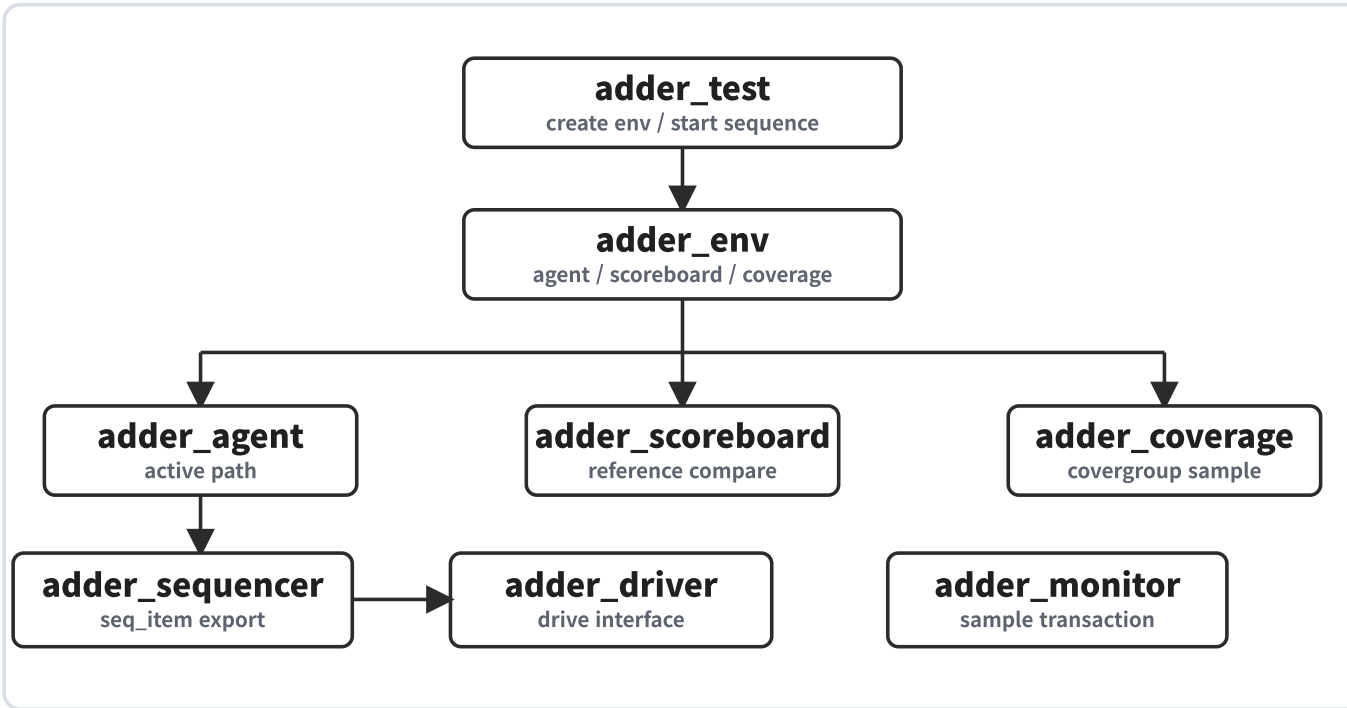
```
module adder #(DATA_WIDTH = 32) (  
    input logic [DATA_WIDTH-1:0] iA,  
    input logic [DATA_WIDTH-1:0] iB,  
    output logic [DATA_WIDTH :0] oY  
);  
    assign oY = iA + iB;  
endmodule
```

Figure 3. Adder UVM Sequence and Combinational Timing

The monitor samples after #1 so the combinational output has settled before scoreboard comparison.



Adder UVM Class 구조



Adder UVM class map

connect_phase

```
agt.monitor.send.connect(scb.recv);  
agt.monitor.send.connect(cov.analysis_export);  
driver.seq_item_port.connect(  
    sequencer.seq_item_export  
);
```

Adder에서 달라지는 점

- clocking block 없이 조합 settle 후 sample
- scoreboard reference model은 상태가 없는 즉시 계산
- coverage는 입력 corner와 carry 발생 여부 중심

Adder Sequence / Coverage

sequence 구성

구간	내용
directed	zero / misc / max 조합 9개
random	50개 transaction
핵심 case	max + max carry 발생

coverage bins

- cp_iA, cp_iB: zero / max / misc
- cp_carry: no_carry / carry
- cross_iA_iB: operand 조합

```
UVM_INFO adder_coverage.sv(45) @ 69000: uvm_test_top.env.cov [ADDER_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 69000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 73
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[ADDER_COV] 1
[ADDER_SCB] 69
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

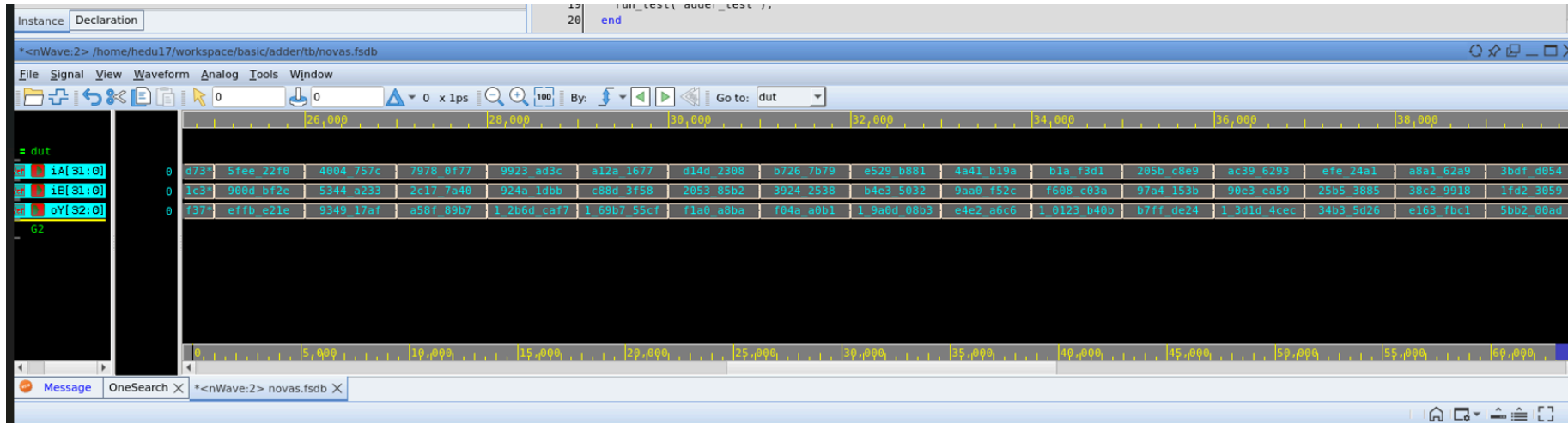
$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 69000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 69000 ps
CPU Time: 0.210 seconds; Data structure size: 0.6Mb
Mon May 11 15:13:59 2026
```


Adder 결과



Adder Verdi waveform

Summary

Hierarchy Modules Groups Asserts Statistics Tests

Avg. Group Score:100.00% U+C:10 U:0 C:10 X:0
Avg. Group Inst. Score:100.00% U+C:10 U:0 C:10 X:0

Group	Score	Instances	U+C	U	C	X
tb_adder_pkg::adder_cov...	100.00%	100.00%	10	0	10	
cp_carry	100.00%		2	0	2	
cp_iA	100.00%		2	0	2	
cp_iB	100.00%		2	0	2	
cross_iA_iB	100.00%		4	0	4	
tb_adder.me.obj.add...	100.00%		10	0	10	

UVM result summary

```
JVM_INFO adder_coverage.sv(45) @ 69000: uvm_test_top.env.cov [ADDER_COV] Functional coverage = 100.00%
JVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 69000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
JVM_INFO : 73
JVM_WARNING : 0
JVM_ERROR : 0
JVM_FATAL : 0

** Report counts by id
[ADDER_COV] 1
[ADDER_SCB] 69
[RNIST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 69000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 69000 ps
CPU Time: 0.210 seconds; Data structure size: 0.6Mb
Mon May 11 15:13:59 2026
```

Coverage result

RAM RTL

동작 조건

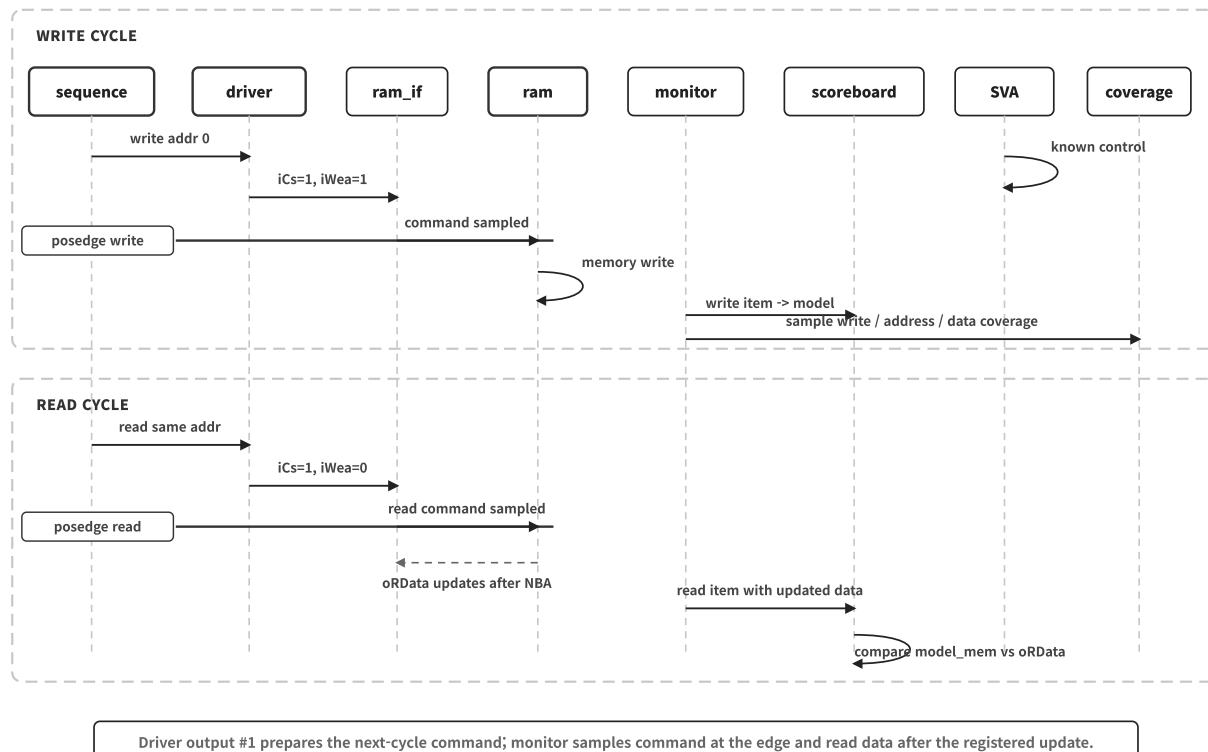
조건	동작
!iRstn	oRData와 memory 초기화
iCs && iWea	mem[iAddr] <= iWData
iCs && !iWea	oRData <= mem[iAddr]

```

always_ff @(posedge iClk or negedge iRstn) begin
    if (!iRstn) begin
        oRData <= '0;
        mem[idx] <= '0;
    end else if (iCs) begin
        if (iWea) mem[iAddr] <= iWData;
        else oRData <= mem[iAddr];
    end
end
    
```

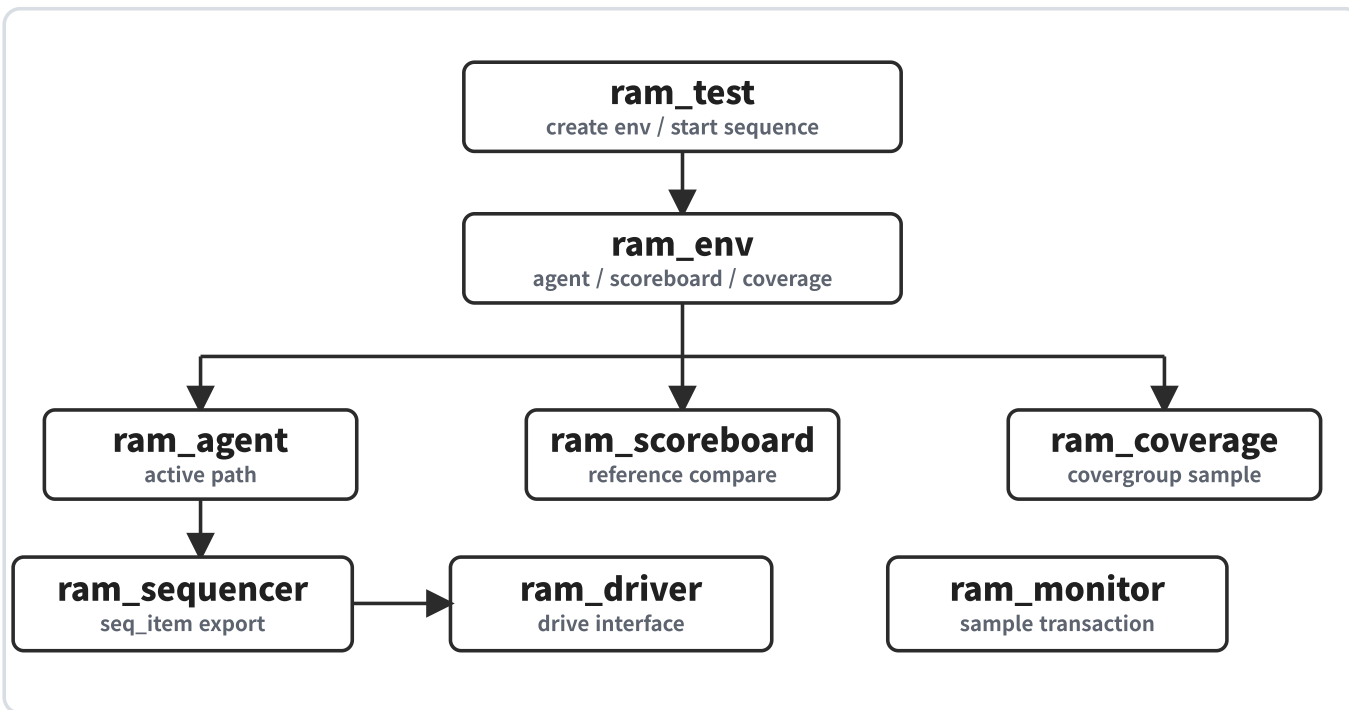
Figure 4. RAM UVM Sequence and Synchronous Timing

Write/read commands are sampled on the active clock edge; read data is checked after oRData settles.



RAM synchronous read/write timing

RAM UVM Class 구조



RAM UVM class map

RAM에서 필요한 처리

- driver는 clocking block으로 command 유지
- monitor는 read command 이후 갱신된 oRData sample
- scoreboard는 reference memory 상태 유지

interface SVA

- selected transaction에서 X/Z control 방지
- write data known check
- reset 중 command idle check

RAM Sequence / Scoreboard / SVA

sequence 구성

구간	내용
directed	first / last / misc address write-read
sweep	0부터 DEPTH-1까지 write/read
random	100개 random access

scoreboard model

```
write: model_mem[addr] = iWData
read : expected = model_mem[addr]
      compare expected with oRData
```

coverage/SVA

- cp_cmd: read / write
- cp_addr: first / last / misc
- cross_cmd_addr + control/data assertion

```
UVM_INFO ram_coverage.sv(50) @ 2815000: uvm_test_top.env.cov [RAM_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvmm-1.2/base/uvmm_report_server.svh(904) @ 2815000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 134
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[RAM_COV] 1
[RAM_SCB] 130
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

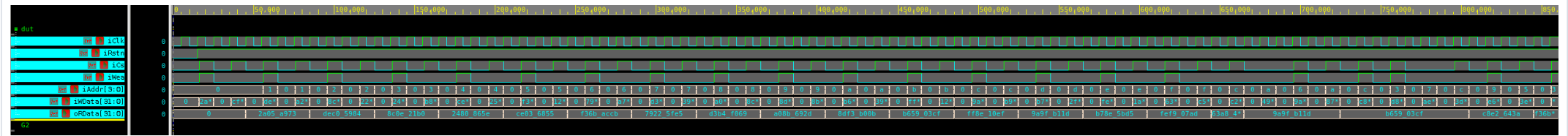
$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvmm-1.2/base/uvmm_root.svh", line 527.
$finish at simulation time 2815000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2815000 ps
CPU Time: 0.230 seconds; Data structure size: 0.6Mb
```

RAM 결과



RAM Verdi waveform

Group	Score	Instances	U+C	U	C	X
tb_ram_pkg::ram_covera...	100.00%	100.00%	12	0	12	
CP cp_addr	100.00%		2	0	2	
CP cp_cmd	100.00%		2	0	2	
CP cp_rdata	100.00%		2	0	2	
CP cp_wdata	100.00%		2	0	2	
CR cross_cmd_addr	100.00%		4	0	4	
tb_ram.me.obj.ram_cg	100.00%		12	0	12	

UVM result summary

```
UVM INFO ram_coverage.sv(50) @ 2815000: uvm_test_top.env.cov [RAM_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 2815000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 134
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0

** Report counts by id
[RAM_COV] 1
[RAM_SCB] 130
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 2815000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2815000 ps
CPU Time: 0.230 seconds; Data structure size: 0.6Mb
```

Coverage result

FIFO RTL

상태 신호

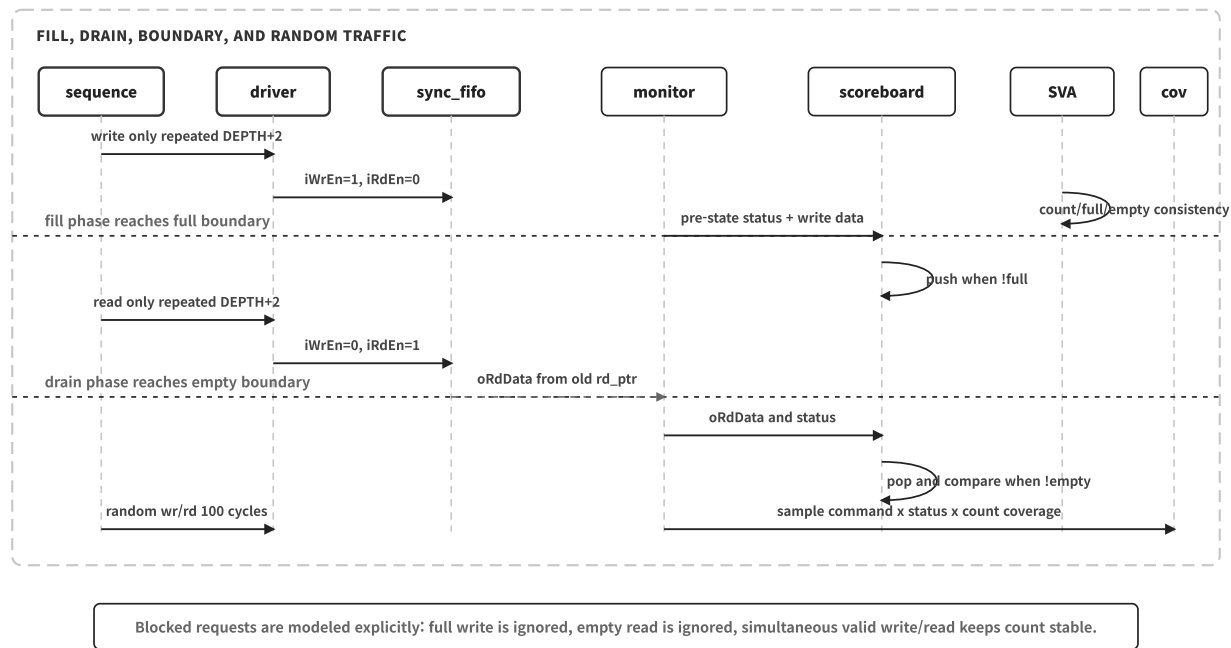
Signal	기준
oFull	oCount == DEPTH
oEmpty	oCount == 0
oRdData	mem[rd_ptr]

```

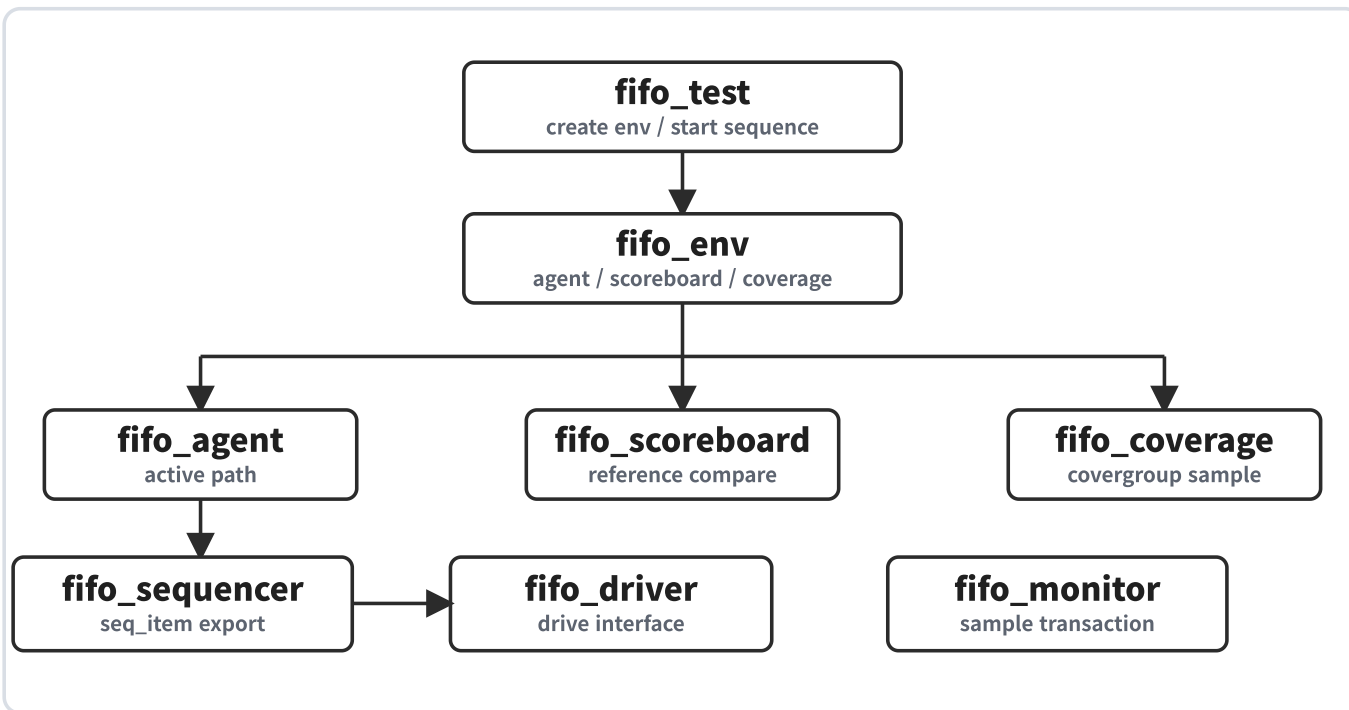
unique case ({iWrEn && !oFull,
             iRdEn && !oEmpty})
  2'b10: write, count++
  2'b01: read,  count--
  2'b11: write and read, count hold
  default: state hold
endcase
    
```

Figure 5. FIFO UVM Sequence and Queue Reference Model

The scoreboard decides wr_fire and rd_fire from pre-state full/empty status sampled by the monitor.



FIFO UVM Class 구조



FIFO UVM class map

FIFO에서 필요한 처리

- read data는 edge 직전 old rd_ptr 기준으로 확인
- scoreboard는 full/empty를 이용해 fire 여부 계산
- count/full/empty consistency assertion 사용

blocked request

- full 상태 write는 push하지 않음
- empty 상태 read는 pop하지 않음
- valid write/read 동시에는 count 유지

FIFO Sequence / Scoreboard / Coverage

sequence 구성

구간	내용
fill	write only DEPTH+2
drain	read only DEPTH+2
mixed	simultaneous write/read + random 100

scoreboard model

```
wr_fire = iWrEn && !oFull
rd_fire = iRdEn && !oEmpty

if (rd_fire) compare(model_q.pop_front(), oRdData)
if (wr_fire) model_q.push_back(iWrData)
```

coverage bins

- cp_cmd: idle/write/read/wr_rd
- cp_count: zero/mid/full
- cross_cmd_status

```
UVM_INFO fifo_coverage.sv(50) @ 2515000: uvm_test_top.env.cov [FIFO_COV] Functional coverage = 100.00%
UVM_INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvmm-1.2/base/uvmm_report_server.svh(904) @ 2515000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 123
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[FIFO_COV] 1
[FIFO_SCB] 119
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

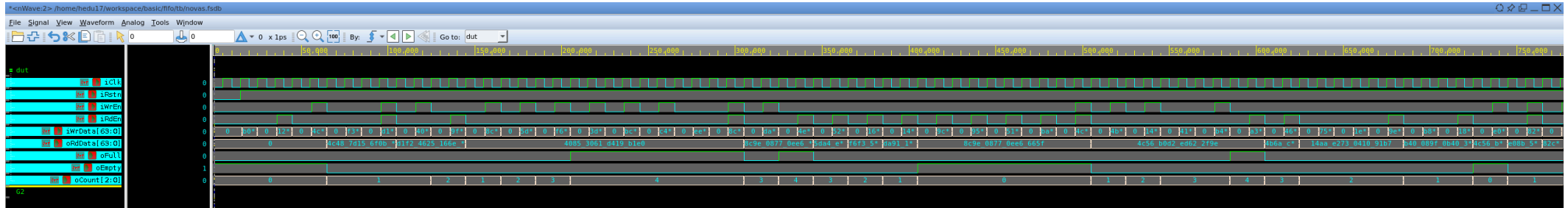
$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvmm-1.2/base/uvmm_root.svh", line 527.
$finish at simulation time 2515000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2515000 ps
CPU Time: 0.220 seconds; Data structure size: 0.6Mb
Mon May 11 15:14:47 2026
```


FIFO 결과



FIFO Verdi waveform

Summary													
Hierarchy Modules Groups Asserts Statistics Tests													
Avg. Group Score:100.00% U+C:23 U:0 C:23 X:0													
Avg. Group Inst. Score:100.00% U+C:23 U:0 C:23 X:0													
Group	Score	Instances	U+C	U	C	X	Goal	Weight	AtLeast	PerInst	Overlap	AutoBini	Missing
tb_fifo_pkg::fifo_covrag...	100.00%	100.00%	23	0	23	0	100%	1	1	1	1	64	64
cp_cmd	100.00%	4	0	4	0	100%	1	1	1	1	1	0	
cp_count	100.00%	3	0	3	0	100%	1	1	1	1	1	0	
cp_empty	100.00%	2	0	2	0	100%	1	1	1	1	1	0	
cp_full	100.00%	2	0	2	0	100%	1	1	1	1	1	0	
cross_cmd_status	100.00%	12	0	12	0	100%	1	1	1	1	1	0	
tb_fifo.me.obj.fifo_cg	100.00%	23	0	23	0	100%	1	1	1	1	1	64	64

UVM result summary

```
UVM INFO fifo_coverage.sv(50) @ 2515000: uvm_test_top.env.cov [FIFO_COV] Functional coverage = 100.00%
UVM INFO /tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_report_server.svh(904) @ 2515000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 123
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[FIFO_COV] 1
[FIFO_SCB] 119
[RNTST] 1
[TEST_DONE] 1
[UVM/RELNOTES] 1

$finish called from file "/tools/synopsys/vcs/W-2024.09-SP1/etc/uvm-1.2/base/uvm_root.svh", line 527.
$finish at simulation time 2515000

-----
VCS Coverage Metrics: during simulation line, cond, FSM, branch, tgl was monitored
-----

Coverage status: End of All Coverages ...

VCS Simulation Report
Time: 2515000 ps
CPU Time: 0.220 seconds; Data structure size: 0.6Mb
Mon May 11 15:14:47 2026
```

Coverage result

블록별 검증 관점 비교

Block	RTL 특성	Monitor timing	Scoreboard model	Coverage 중심
Adder	조합 회로	입력 drive 후 #1 sample	expected = iA + iB	operand corner, carry
RAM	synchronous single-port memory	command edge sample, read data #1 sample	model_mem[DEPTH]	cmd, address, data
FIFO	pointer/count 기반 queue	edge 직전 status/data sample	model_q[\$]	cmd, full/empty, count

조합 회로

- 상태 모델 불필요
- 출력 폭과 carry 확인

Memory

- write/read 순서 모델링
- registered read timing 반영

Queue

- data order와 status 동시 검증
- blocked request 처리

결론

최종 정리

Adder, RAM, FIFO는 동일한 UVM testbench 골격을 공유하지만, 실제 검증 품질은 DUT별 timing rule, reference model, coverage 목표를 분리했을 때 확보된다.

구조 재사용

- test / env / agent / sequence 구조를 세 DUT에 동일하게 적용
- factory 기반 class 생성 흐름과 connect 구조 확인

모델링 차이

- Adder는 즉시 계산 reference model 사용
- RAM은 model_mem, FIFO는 model_q로 상태 유지

검증 증거

- Verdi waveform으로 pin-level timing 확인
- UVM report와 coverage 화면으로 scenario 수행 결과 확인

구분	이번 실습에서 확인한 점	추가 개선 방향
Sequence	directed case와 random case를 함께 구성	constraint 분리, corner regression 확대
Scoreboard	DUT 특성별 reference model을 분리	mismatch log format과 debug trace 정리
Coverage / SVA	기능 coverage와 protocol assertion을 함께 사용	covergroup 목표값과 assertion 항목 문서화